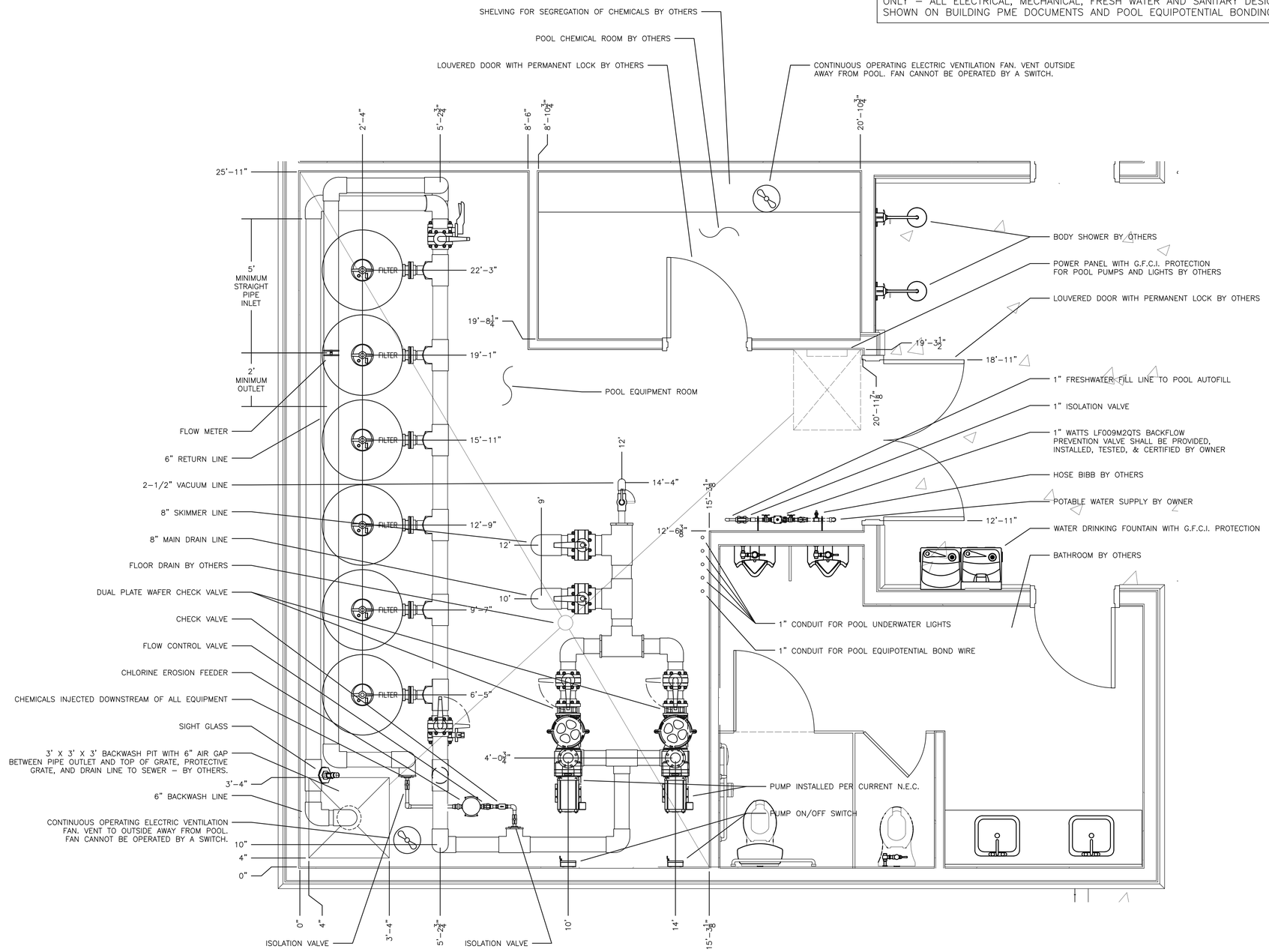
POOL ELECTRICAL REQUIREMENTS

QTY	EQUIPMENT	ELECTRICAL	NOTES (WORK BY OTHERS)
2	7.5 HP PUMP, FILTRATION	230 VAC, 30.4 AMPS, SINGLE PHASE	RUN POWER TO DISCONNECT IN EQUIPMENT ROOM
9	UNDERWATER POOL LIGHT, LED	55 WATT, 120 VAC, SINGLE PHASE, 60 CYCLE	POWER FROM POOL EQUIPMENT ROOM GFCI CIRCUITS TO JUNCTION BOXES ON POOL DECK
1	POOL EQUIPOTENTIAL BONDING GRID	MINIMUM 8 AWG SOLID COPPER BONDING WIRE	EQUIPOTENTIAL BOND GRID AT POOL PERIMETER AND CONTINUOUS #8 COPPER WIRE TO POOL EQUIPMENT ROOM

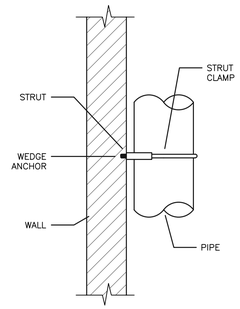
NOTE: WE DO NOT ALWAYS KNOW THE POWER SUPPLY COMING INTO THE POOL EQUIPMENT ROOM. IN THE EVENT 208–230V OR 230V SINGLE PHASE PUMPS ARE SPECIFIED ON THIS PROJECT, **ELECTRICAL ENGINEERS PLEASE NOTE: IF THE SUPPLY TO THE POOL EQUIPMENT ROOM IS 208 VOLTS, "BUCK BOOSTERS" ARE REQUIRED TO ENSURE THE PUMP HAS THE CORRECT VOLTAGE OR MOTOR FAILURE MAY OCCUR.

NOTE:
ALL SCHEDULE 40 PVC PRESSURE RATED PIPING UNLESS NOTED OTHERWISE.
ONLY SDR 26 (CLASS 160) AND SDR 21 (CLASS 200) PVC PIPE MEETING ASTM STANDARD D1785 OR D2241 ARE ACCEPTABLE IN SIZES 12" & SMALLER. NO HEAT BENDING OF PVC PIPE IS ALLOWED. CPVC PIPE SHALL BE USED FOR CONNECTING TO HEATERS.

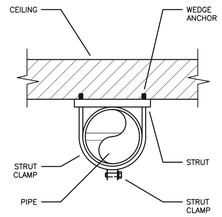
ELECTRICAL AND MECHANICAL NOTES ARE SHOWN FOR COORDINATION PURPOSES ONLY – ALL ELECTRICAL, MECHANICAL, FRESH WATER AND SANITARY DESIGN TO BE SHOWN ON BUILDING PME DOCUMENTS AND POOL EQUIPOTENTIAL BONDING PLAN.



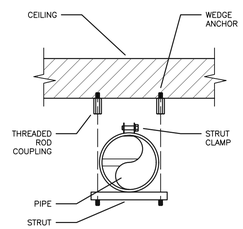
1 / SP-11 SCALE: 1/8" = 1'-0"



2 / SP-11 SCALE: NTS



3 / SP-11 SCALE: NTS



4 / SP-11 SCALE: NTS

0' 2' 4' 6'
1/2"=1'-0"

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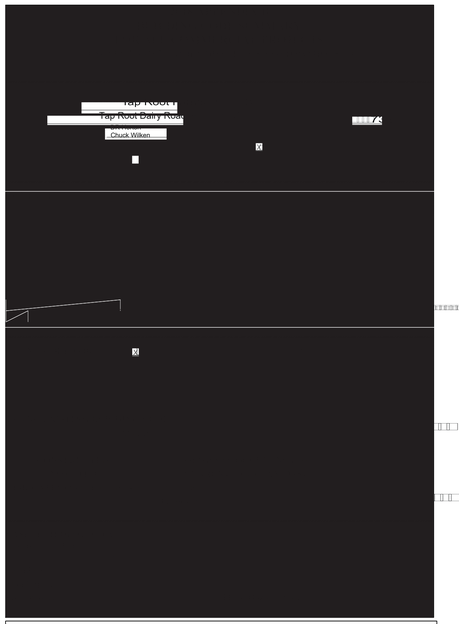
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SEAL:

Drawing Name: PLUMBING LAYOUT

PROJECT NO: 015

DESIGN BY:	PAUL	DRAWING NO.	EM-0
SCALE:	N/A		
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00



Gross Building Area:

FLOOR	EXISTING (SQ FT)	NEW (SQ FT)	RENO/ALTER (SQ.FT)	SUB-TOTAL
6th Floor				
5th Floor				
4th Floor				
3rd Floor				
2nd Floor				
Mezzanine				
1st Floor			8456	
Basement			11819	
TOTAL			20,275	

ALLOWABLE AREA

Primary Occupancy Classification: SELECT ONE

Assembly ☐ A-1 ☐ A-2 ☐ A-3 ☐ A-4 ☐ A-5
Business ☐
Educational ☐
Factory ☐ F-1 Moderate ☐ F-2 Low
Hazardous ☐ H-1 Detonate ☐ H-2 Deflagrate ☐ H-3 Combust ☐ H-4 Health ☐ H-5 HPM
Institutional ☐ I-1 Condition ☐ 1 ☐ 2
☐ 1-2 Condition ☐ 1 ☐ 2
☐ 1-3 Condition ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
☐ 1-4
Mercantile ☐
Residential ☐ R-1 ☐ R-2 ☐ R-3 ☐ R-4
Storage ☐ S-1 Moderate ☐ S-2 Low ☐ High-piled
☐ Parking Garage ☐ Open ☐ Enclosed ☐ Repair Garage
Utility and Miscellaneous ☐

Accessory Occupancy Classification(s): ____
Incidental Uses (Table 509): ____
Special Uses (Chapter 4 – List Code Sections) ____
Special Provisions: (Chapter 5 – List Code Sections): ____
Mixed Occupancy: ☐ No ☐ Yes Separation: ____ Hr. Exception: ____

☐ Non-Separated Use (508.3)
The required type of construction for the building shall be determined by applying the height and area limitations for each of the applicable occupancies to the entire building. The most restrictive type of construction, so determined, shall apply to the entire building.

☐ Separated Use (508.4) -
See below for area calculations for each story, the area of the occupancy shall be such that the sum of the ratios of the actual floor area of each use divided by the allowable floor area for each use shall not exceed 1.

$$\frac{\text{Actual Area of Occupancy A}}{\text{Allowable Area of Occupancy A}} + \frac{\text{Actual Area of Occupancy B}}{\text{Allowable Area of Occupancy B}} \leq 1$$

_____ + _____ + ... = _____ ≤ 1.00

STORY NO.	DESCRIPTION AND USE	(A) BLDG AREA PER STORY (ACTUAL)	(B) TABLE 506.2[1] AREA	(C) AREA FOR FRONTAGE INCREASE[1,2]	(D) ALLOWABLE AREA PER STORY OR UNLIMITED[2,3]

[1] Frontage area increases from Section 506.3 are computed thus:
a. Perimeter which fronts a public way or open space having 20 feet minimum width = _____ (F)
b. Total Building Perimeter = _____ (P)
c. Ratio (F/P) = _____ (F/P)
d. W = Minimum width of public way = _____ (W)
e. Percent of frontage increase $I_f = 100 [F/P - 0.25] \times W/30$ = _____ (%)
[2] Unlimited area applicable under conditions of Section 507.
[3] Maximum Building Area = total number of stories in the building x D (maximum 3 stories) (506.2).
[4] The maximum area of open parking garages must comply with Table 406.5.4
[5] Frontage increase is based on the unsprinklered area value in Table 506.2.

STORY NO.	DESCRIPTION AND USE	(A) BLDG AREA PER STORY (ACTUAL)	(B) TABLE 506.2[4] AREA	(C) AREA FOR FRONTAGE INCREASE[1,5]	(D) ALLOWABLE AREA PER STORY OR UNLIMITED[2,3]

Frontage area increases from Section 506.3 are computed thus:
a. Perimeter which fronts a public way or open space having 20 feet minimum width = _____ (F)
b. Total Building Perimeter = _____ (P)
c. Ratio (F/P) = _____ (F/P)
d. W = Minimum width of public way = _____ (W)
e. Percent of frontage increase $I_f = 100 [F/P - 0.25] \times W/30$ = _____ (%)

[1] Unlimited area applicable under conditions of Section 507.
[2] Maximum Building Area = total number of stories in the building x D (maximum 3 stories) (506.2).
[3] The maximum area of open parking garages must comply with Table 406.5.4
[4] Frontage increase is based on the unsprinklered area value in Table 506.2.

FIRE PROTECTION REQUIREMENTS

BUILDING ELEMENT	FIRE SEPARATION DISTANCE (FEET)	RATING REQ'D	RATING PROVIDED (W/ * REDUCTION)	DETAIL # AND SHEET #	DESIGN # FOR RATED ASSEMBLY	DESIGN # FOR RATED PENETRATION	DESIGN # FOR RATED JOINTS
Structural Frame, including columns, girders, trusses							
Bearing Walls							
Exterior							
North							
East							
West							
South							
Interior							
Nonbearing Walls and Partitions							
Exterior walls							
North							
East							
West							
South							
Interior walls and partitions							
Floor Construction Including supporting beams and joists							
Floor Ceiling Assembly							
Column Supporting Floors							
Roof Construction, including supporting beams and joists							
Roof Ceiling Assembly							
Column Supporting Roof							
Shaft Enclosures - Exit							
Shaft Enclosures - Other							
Corridor Separation							
Occupancy/Fire Barrier Separation							
Party/Fire Wall Separation							
Smoke Barrier Separation							
Smoke Partition							
Tenant/Dwelling Unit/ Sleeping Unit Separation							
Incidental Use Separation							

* Indicate section number permitting reduction

PERCENTAGE OF WALL OPENING CALCULATIONS

FIRE SEPARATION DISTANCE (FEET FROM PERPERTY LINES	DEGREES OF OPENINGS PROTECTION (TABLE 705.8)	ALLOWABLE AREA (%)	ACTUAL SHOWN ON PLANS (%)

LIFE SAFETY SYSTEM REQUIREMENTS

Emergency Lighting: ☐ No ☐ Yes
Exit Signs: ☐ No ☐ Yes
Fire Alarm: ☐ No ☐ Yes
Smoke Detection Systems: ☐ No ☐ Yes ☐ Partial _____
Carbon Monoxide Detection: ☐ No ☐ Yes

LIFE SAFETY PLAN REQUIREMENTS

Life Safety Plan Sheet #: N/A

☐ Fire and/or smoke rated wall locations (Chapter 7)
☐ Assumed and real property line locations (if not on the site plan)
☐ Exterior wall opening area with respect to distance to assumed property lines (705.8)
☐ Occupancy types for each area as it relates to occupant load calculation (Table 1004.1.2)
☐ Occupant loads for each area
☐ Exit access travel distances (1017)
☐ Common path of travel distances (1006.2.1 & 2006.3.2(1))
☐ Dead end lengths (1020.4)
☐ Clear exit widths for each exit door
☐ Maximum calculated occupant load capacity each exit door can accommodate based on egress width (1005.3)
☐ Actual occupant load for each exit door
☐ A separate schematic plan indicating where fire rated floor/ceiling and/or roof structure is provided for purposes of occupancy separation and supporting construction for a fire barrier/fire partition/smoke barrier.
☐ Location of doors with panic hardware (1010.1.10)
☐ Location of doors with delayed egress locks and the amount of delay (1010.1.9.7)
☐ Location of doors with electromagnetic egress locks (1010.1.9.9)
☐ Location of doors equipped with hold-open devices
☐ Location of emergency escape windows (1030)
☐ The square footage of each fire area (202)
☐ The square footage of each smoke compartment for Occupancy Classification I-2 (407.5)
☐ Note any code exceptions or table notes that may have been utilized regarding the items above

Section/Table/Note	Title

ACCESSIBLE DWELLING UNITS
(SECTION 1107)

TOTAL UNITS	ACCESSIBLE UNITS REQUIRED	ACCESSIBLE UNITS PROVIDED	TYPE A UNITS REQUIRED	TYPE A UNITS PROVIDED	TYPE B UNITS REQUIRED	TYPE B UNITS PROVIDED	TOTAL ACCESSIBLE UNITS PROVIDED

ACCESSIBLE PARKING
(SECTION 1106)

LOT OR PARKING AREA	TOTAL # OF PARKING SPACES REQUIRED	TOTAL # OF PARKING SPACES PROVIDED	# OF ACCESSIBLE SPACES PROVIDED REGULAR WITH 5' ACCESS AISLE	VAN SPACES WITH 132" ACCESS AISLE	VAN SPACES WITH 8' ACCESS AISLE	TOTAL # ACCESSIBLE PROVIDED
TOTAL						

PLUMBING FIXTURE REQUIREMENTS
(TABLE 2902.1)

USE		WATERCLOSETS MALE	WATERCLOSETS FEMALE	WATERCLOSETS UNISEX	URINALS	LAVATORIES MALE	LAVATORIES FEMALE	LAVATORIES UNISEX	SHOWERS / TUBS	DRINKING FOUNTAINS REGULAR	DRINKING FOUNTAINS ACCESSIBLE
SPACE	EXIST'G										
	NEW	2	6	0	2	2	3	0	2	1	1
	REQ'D	2	6	0	2	2	3		2	1	1

SPECIAL APPROVALS

Special approval: (Local Jurisdiction, Department of Insurance, SCO, DPI, DHHS, ICC, etc., describe below)

ENERGY SUMMARY

ENERGY REQUIREMENTS:
The following data shall be considered minimum and any special attribute required to meet the **North Carolina Energy Conservation Code** shall also be provided. Each Designer shall furnish the required portions of the project information for the plan data sheet. If performance method, state the annual energy cost for the standard reference design vs annual energy cost for the proposed design.

Existing building envelope complies with code: ☐ No ☐ Yes (The remainder of this section is not applicable)

Exempt Building: ☐ No ☐ Yes (Provide Code or Statutory reference): _____

Climate Zone: ☐ 3A ☐ 4A ☐ 5A

Method of Compliance: Energy Code ☐ Performance ☐ Prescriptive
ASHRAE 90.1 ☐ Performance ☐ Prescriptive
(If "Other" specify source here) _____

THERMAL ENVELOPE (Prescriptive method only)

Roof/ceiling Assembly (each assembly)
Description of assembly: _____
U-Value of total assembly: _____
R-Value of insulation: _____
Skylights in each assembly: _____
U-Value of skylight: _____
Total square footage of skylights in each assembly: _____

Exterior Walls (each assembly)
Description of assembly: _____
U-Value of total assembly: _____
R-Value of insulation: _____
Openings (windows or doors with glazing)
U-Value of assembly: _____
Solar heat gain coefficient: _____
Projection factor: _____
Door R-Values: _____

Walls below grade (each assembly)
Description of assembly: _____
U-Value of total assembly: _____
R-Value of insulation: _____

Floors over unconditioned space (each assembly)
Description of assembly: _____
U-Value of total assembly: _____
R-Value of insulation: _____

Floors slab on grade
Description of assembly: _____
U-Value of total assembly: _____
R-Value of insulation: _____
Horizontal/Vertical requirement: _____
Slab Heated: _____

2018 APPENDIX B
BUILDING CODE SUMMARY FOR ALL COMMERCIAL PROJECTS
STRUCTURAL DESIGN
(PROVIDE ON THE STRUCTURAL SHEETS IF APPLICABLE)

DESIGN LOADS:

Importance Factors: Snow (I_s) _____
Seismic (I_E) _____

Live Loads: Roof _____ psf
Mezzanine _____ psf
Floor _____ psf

Ground Snow Load: _____ psf

Wind Load: Ultimate Wind Speed _____ mph (ASCE-7)
Exposure Category _____

SEISMIC DESIGN CATEGORY: ☐ A ☐ B ☐ C ☐ D
Provide the following Seismic Design Parameters:
Occupancy Category (Table 1604.5) ☐ I ☐ II ☐ III ☐ IV
Spectral Response Acceleration S_s _____ %g S_1 _____ %g
Site Classification (ASCE 7) ☐ A ☐ B ☐ C ☐ D ☐ E ☐ F
Data Source: ☐ Field Test ☐ Presumptive ☐ Historical Data
Basic structural system ☐ Bearing Wall ☐ Dual w/Special Moment Frame
☐ Building Frame ☐ Dual w/Intermediate R/C or Special Steel
☐ Moment Frame ☐ Inverted Pendulum
Analysis Procedure: ☐ Simplified ☐ Equivalent Lateral Force ☐ Dynamic
Architectural, Mechanical, Components anchored? ☐ Yes ☐ No

LATERAL DESIGN CONTROL: Earthquake ☐ Wind ☐

SOIL BEARING CAPACITIES:
Field Test (provide copy of test report) _____ psf
Presumptive Bearing capacity _____ psf
Pile size, type, and capacity _____

2018 APPENDIX B
BUILDING CODE SUMMARY FOR ALL COMMERCIAL PROJECTS
MECHANICAL DESIGN
(PROVIDE ON THE MECHANICL SHEETS IF APPLICABLE)

MECHANICAL SUMMARY

MECHANICAL SYSTEMS, SERVICE SYSTEMS AND EQUIPMENT

Thermal Zone
winter dry bulb: _____
summer dry bulb: _____

Interior design conditions
winter dry bulb: _____
summer dry bulb: _____
relative humidity: _____

Building heating load: _____

Building cooling load: _____

Mechanical Spacing Conditioning System
Unitary
description of unit: _____
heating efficiency: _____
cooling efficiency: _____
size category of unit: _____
Boiler
Size category. If oversized, state reason.: _____
Chiller
Size category. If oversized, state reason.: _____

List equipment efficiencies: _____

2018 APPENDIX B
BUILDING CODE SUMMARY FOR ALL COMMERCIAL PROJECTS
ELECTRICAL DESIGN
(PROVIDE ON THE ELECTRICAL SHEETS IF APPLICABLE)

ELECTRICAL SUMMARY

ELECTRICAL SYSTEM AND EQUIPMENT

Method of Compliance: Energy Code: ☐ Prescriptive ☐ Performance
ASHRAE 90.1: ☐ Prescriptive ☐ Performance

Lighting schedule (each fixture type)
lamp type required in fixture
number of lamps in fixture
ballast type used in the fixture
number of ballasts in fixture
total wattage per fixture
total interior wattage specified vs. allowed (whole building or space by space)
total exterior wattage specified vs. allowed

Additional Efficiency Package Options
(When using the 2018 NCECC; not required for ASHRAE 90.1)
☐ C406.2 More Efficient Mechanical Equipment
☐ C406.3 Reduced Lighting Power Density
☐ C406.4 Enhanced Digital Lighting Controls
☐ C406.5 On-Site Renewable Energy
☐ C406.6 Dedicated Outdoor Air System
☐ C406.7 Reduced Energy Use in Service Water Heating

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TAP ROOT FARMS- MAIN
ENTRY FLETCHER, NC 28732

NO.	EMISSIONS & REVISIONS	DATES
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SEAL:

Drawing Name:

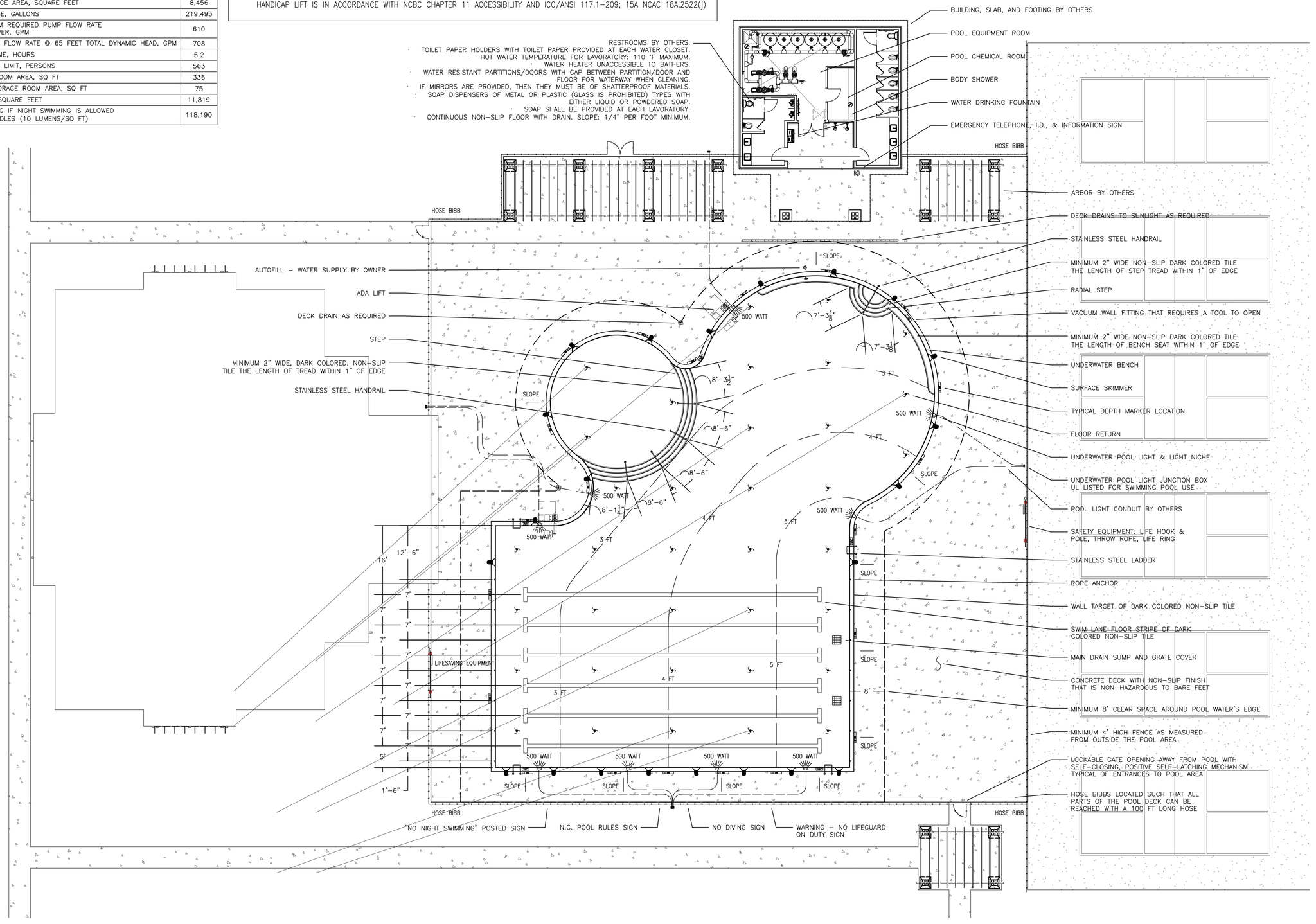
SITE PLAN

PROJECT NO: 015

DESIGN BY:	PAUL	DRAWING NO.	EM-1
SCALE:	1" = 10'-0"		
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00

OUTDOOR POOL DATA	
NORTH CAROLINA POOL TYPE	PUBLIC
PERIMETER AT WATER'S EDGE, FT	408
WATER SURFACE AREA, SQUARE FEET	8,456
WATER VOLUME, GALLONS	219,493
POOL MINIMUM REQUIRED PUMP FLOW RATE 6 HR TURNOVER, GPM	610
DESIGN PUMP FLOW RATE @ 65 FEET TOTAL DYNAMIC HEAD, GPM	708
TURNOVER TIME, HOURS	5.2
POOL BATHER LIMIT, PERSONS	563
EQUIPMENT ROOM AREA, SQ FT	336
CHEMICAL STORAGE ROOM AREA, SQ FT	75
DECK AREA, SQUARE FEET	11,819
DECK LIGHTING IF NIGHT SWIMMING IS ALLOWED 10 FOOT CANDLES (10 LUMENS/SQ FT)	118,190

NOTE:
POOL TO BE CONSTRUCTED IN ACCORDANCE WITH 15A NCAC 18A.2500 AND ASME/ANSI A112.19.8-2007
SUCTION OUTLETS WILL BE INSTALLED IN ACCORDANCE WITH 3109.5 ENTRAPMENT AVOIDANCE AND ANSI/APSP/ICC-16 2017
HANDICAP LIFT IS IN ACCORDANCE WITH NCBC CHAPTER 11 ACCESSIBILITY AND ICC/ANSI 117.1-209; 15A NCAC 18A.2522(j)



1 / SP-1 SCALE: 3/32" = 1'-0"

0' 4' 8' 12' 24' 30'
3/32" = 1'-0"

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SEAL:

Drawing Name: PLUMBING LAYOUT

PROJECT NO:	015
DESIGN BY: PAUL	DRAWING NO. EM-2
SCALE: N/A	
DRAWN BY: IVAN	
VERIFY BY: PAUL	
START DATE: 16 AUG 2023	REVISION # 00

680.26 Equipotential Bonding

680.26 (A) Performance. The equipotential bonding required by this section shall be installed to reduce voltage gradients in the pool area.

680.26 (B) Bonded Parts. The parts specified in 680.26(B)(1) through (B)(7) shall be bonded together using solid copper conductors, insulated covered, or bare, not smaller than 8 AWG or with rigid metal conduit of brass or other identified corrosion-resistant metal. Connections to bonded parts shall be made in accordance with 250.8 (see note following). An 8 AWG or larger solid copper bonding conductor provided to reduce voltage gradients in the pool area shall not be required to be extended or attached to remote panel boards, service equipment, or electrodes.

NOTE: Article 250.8 specifies equipment grounding conductors, grounding electrode conductors, and bonding jumpers shall be connected by pressure connectors listed as grounding and bonding equipment.

680.26(B)(1) Conductive Pool Shells. Bonding to conductive pool shells shall be provided as specified in 680.26(B)(1)(a) or (B)(1)(b). Poured concrete, pneumatically applied or sprayed concrete and concrete block with painted or plastered coatings shall all be considered conductive materials due to water permeability and porosity. Vinyl liners and fiberglass composite shells shall be considered to be non-conductive materials.
(a) Structural Reinforcing Steel. Un-encapsulated structural reinforcing steel shall be bonded together by steel tie wires or the equivalent. Where structural reinforcing steel is encapsulated in a nonconductive compound, a copper conductor grid shall be installed in accordance with 680.26(B)(1)(b).
(b) Copper Conductor Grid. A copper conductor grid shall be provided and shall comply with (b)(1) through (b)(4).
(1) Be constructed of minimum 8 AWG bare solid copper conductors bonded to each other at all points of crossing. The bonding shall be in accordance with 250.8 or other approved means.
(2) Conform to the contour of the pool.
(3) Be arranged in a 300-mm (12-in.) by 300-mm (12-in.) network of conductors in a uniformly spaced perpendicular grid pattern with a tolerance of 100-mm (4 in.).
(4) Be secured within or under the pool no more than 150-mm (6 in.) from the outer contour of the pool shell.

680.26(B)(2) Perimeter Surfaces. The perimeter surface shall extend for 1 m (3 ft) horizontally beyond the inside walls of the pool and shall include unpaved surfaces, as well as poured concrete surfaces and other types of paving. Perimeter surfaces less than 1 m (3 ft) separated by a permanent wall or building 1.5 m (5 ft) in height or more shall require equipotential bonding on the pool side of the permanent wall or building. Bonding to perimeter surfaces shall be provided as specified in 680.26(B)(2)(a) or (2)(b) and shall be attached to the pool reinforcing steel or copper conductor grid at a minimum of four (4) points uniformly spaced around the perimeter of the pool. For nonconductive pool shells, bonding at four points shall not be required.
(a) Structural Reinforcing Steel. Structural reinforcing steel shall be bonded in accordance with 680.26(B)(1)(a).
(b) Alternate Means. Where structural reinforcing steel is not available or is encapsulated in a non-conductive compound, a copper conductor(s) shall (1) At least one minimum 8 AWG bare solid copper conductor shall be provided.
(2) The conductors shall follow the contour of the perimeter surface.
(3) Only listed splices shall be permitted.
(4) The required conductor shall be 450 mm to 600 mm (18 in. to 24 in.) from the inside walls of the pool.
(5) The required conductor shall be secured within or under the perimeter surface 100 mm to 150 mm (4 in. to 6 in.) below the subgrade.
680.26(B)(3) Metallic Components. All metallic parts of the pool structure., including reinforcing metal not addressed in 680.26(B)(1)(a), shall be bonded. Where reinforcing steel is encapsulated with a nonconductive compound, the reinforcing steel shall not be required to be bonded.

680.26(B)(4) Underwater Lighting. All metal forming shells and mounting brackets of no-niche luminaries shall be bonded.
Exception: Listed low-voltage lighting systems with non-metallic forming shells shall not require bonding.

680.26(B)(5) Metal Fittings. All metal fittings within or attached to the pool structure shall be bonded, isolated parts that are not over 100 mm (4 in.) in any dimension and do not penetrate into the pool structure more than 25 mm (1 in.) shall not require bonding.

680.26(B)(6) Electrical Equipment. Metal parts of electrical equipment associated with the pool water circulating system, including pump motors and metal parts of equipment associated with pool covers, including electric motors, shall be bonded.

Exception: Metal parts of listed equipment incorporating an approved system of double insulation shall not be bonded.

(a) Double-insulated Water Pump Motors. Where a double-insulated water pump motor is installed under the provisions of this rule, a solid 8 AWG copper conductor of sufficient length to make a bonding connection to a replacement motor shall be extended from the bonding grid to an accessible point in the vicinity of the pool pump motor. Where there is no connection for the premises, this bonding conductor shall be connected to the equipment grounding conductor of the motor circuit.
(b) Pool Water Heaters. For pool water heaters rated at more than 50 amperes and having specific instructions regarding bonding and grounding, only those parts designated to be bonded shall be bonded and only those parts designated to be grounded shall be grounded.

680.26(B)(7) Fixed Metal Parts. All fixed metal parts shall be bonded including, but not limited to, metal-sheathed cables and raceways, metal piping, metal awnings, metal fences, and metal door and window frames.

Exception No. 1: Those separated from the pool by a permanent barrier that prevents contact by a person shall not be required to be bonded.

Exception No. 2: Those greater than 1.5m (5 ft) horizontally of the inside walls of the pool shall not be required to be bonded.

Exception No. 3: Those greater than 3.7m (12 ft) measured vertically above the maximum water level of the pool, or as measured vertically above any observation stands, towers, or platforms, or any diving structures, shall not be required to be bonded.

680.26(C) Pool Water. An intentional bond of a minimum conductive surface area of 5800 mm2 (9 in.2) shall be installed in contact with the pool water. This bond shall be permitted to consist of parts that are required to be bonded in 680.26(B).

POOL SHELL CONCRETE REQUIREMENTS:

STANDARDS:

ACI 506.2 SPECIFICATIONS FOR MATERIALS, PROPORTIONING, AND APPLICATION OF GUNITE

ACI 506.3R GUIDE TO CERTIFICATION OF GUNITE NOZZLEMEN

GENERAL:
1. GUNITE WORK SHALL BE PERFORMED BY A FIRM OR COMPANY REGULARLY ENGAGED IN THE BUSINESS OF APPLYING GUNITE MATERIALS, USING NOZZLE OPERATORS AND WORKERS SKILLED AND EXPERIENCED IN THE TYPE OF WORK SPECIFIED.
2. GUNITE SUPERVISOR SHALL HAVE NOT LESS THAN TWO YEARS EXPERIENCE AS A GUNITE NOZZLE OPERATOR.
3. NOZZLE OPERATOR SHALL HAVE NOT LESS THAN ONE YEAR EXPERIENCE AND, UPON REQUEST OF THE ENGINEER, SHALL DEMONSTRATE ABILITY TO PROPERLY PLACE GUNITE.
4. PROOF OF COMPLIANCE WITH ACI 506.3R SHALL BE FURNISHED FOR EACH NOZZLE OPERATOR.
5. GUNITE SHALL NOT BE PLACED DURING INCLEMENT OR WINDY WEATHER.

MIX DESIGN
1. DESIGN OF PREDAMPENED GUNITE DRY MIX, INCLUDING RECOMMENDED AMOUNTS OF ADMIXTURE AND WATER TO BE USED, SHALL BE OBTAINED BY THE CONTRACTOR FROM A QUALIFIED INDEPENDENT TESTING LABORATORY OR AGENCY, OR FROM A MILL OR READY-MIX PLANT, PROPERLY EQUIPPED TO DESIGN GUNITE MIXES. THE LABORATORY, AGENCY, MILL OR READY- MIX PLANT SHALL MEET THE APPLICABLE REQUIREMENTS OF ASTM E329, AND SHALL MEET WITH APPROVAL OF THE ENGINEER. THE MIX DESIGN SHALL BE CERTIFIED AND SIGNED BY A PROFESSIONAL ENGINEER.
2. UPON RECEIPT OF ACCEPTABLE GUNITE MIX DESIGN AND TEST RESULTS FROM THE PRE-APPROVED INDEPENDENT TESTING LABORATORY, AGENCY, MILL, OR READY-MIX PLANT, CONFORMING WITH SPECIFIED REQUIREMENTS, THE CONTRACTOR SHALL SUBMIT THE ACCEPTED MIX DESIGN TO THE ENGINEER FOR REVIEW PRIOR TO PLACING ANY GUNITE.

GUNITE MIX SHALL CONFORM WITH THE FOLLOWING REQUIREMENTS:
1. PORTLAND CEMENT OF AMERICAN MANUFACTURE SHALL MEET A.S.T.M. C-150-67 'STANDARD SPECIFICATIONS FOR PORTLAND CEMENT'. TYPE I PORTLAND SHALL BE USED UNLESS SPECIFIED IN THE DETAILED SPECIFICATIONS.
2. FINE AGGREGATES SHALL CONSIST OF WASHED SAND AND SHALL BE HARD, DENSE, DURABLE, CLEAN, SHARP AND GRADED EVENLY FROM FINE TO COARSE IN ACCORDANCE WITH THE STANDARD SPECIFICATIONS FOR CONCRETE AGGREGATES, A.S.T.M. DESIGNATION: C33-67. IT SHALL BE FREE FROM ORGANIC MATTER AND SHALL NOT CONTAIN MORE THAN 5% BY WEIGHT OF DELETERIOUS SUBSTANCES.
3. PROPORTION OF CEMENT TO SAND AGGREGATE SHALL BE AS REQUIRED TO ACHIEVE THE INDICATED OR SPECIFIED STRENGTH.
4. COMPRESSIVE STRENGTH OF GUNITE SHALL BE NOT LESS THAN 4000 PSI 28-DAY COMPRESSIVE STRENGTH.
5. WATER USED FOR HYDRATION AT THE NOZZLE SHALL BE FIT FOR DRINKING AND SHALL BE MAINTAINED AT A UNIFORM PRESSURE WHICH SHALL BE AT LEAST 15 POUNDS PER SQUARE INCH ABOVE AIR PRESSURE AT THE NOZZLE.
6. WATER CONTENT AT TIME OF DISCHARGE FROM NOZZLE SHALL NOT EXCEED AMOUNT REQUIRED TO ACHIEVE THE MAXIMUM PERMITTED SLUMP.

GUNNING/NOZZLE OPERATION:
1. BUILD EACH LAYER BY MAKING SEVERAL PASSES OVER THE WORKING AREA. THICKNESS OF EACH LAYER SHALL BE GOVERNED BY THE REQUIREMENT THAT SAGGING OF GUNITE SHALL NOT OCCUR. MAINTAIN TOP SURFACE OF THICK LAYERS AT 45 DEGREE SLOPE. EACH LAYER TO BE COVERED BY A SUCCEEDING LAYER SHALL BE ALLOWED TO TAKE ITS INITIAL SET.
2. LAITANCE, LOOSE MATERIAL, AND REBOUND SHALL BE REMOVED BY AIR-JETTING. LAITANCE THAT HAS TAKEN A FINAL SET SHALL BE REMOVED BY SANDBLASTING AND THE SURFACE CLEANED WITH AIR-WATER JET. ALL LAYERS TO BE SHOT SHALL BE DAMP.
3. UNLESS OTHERWISE PERMITTED, BEGIN APPLICATION AT THE LOWEST ELEVATION.
4. DO NOT TROWEL OR FINISH INITIAL LAYERS IN ANY WAY.
5. REBOUND: ANY REBOUND OR ACCUMULATED LOOSE AGGREGATE SHALL BE REMOVED FROM THE SURFACE TO BE COVERED PRIOR TO PLACING SUCCEEDING LAYERS. REBOUND SHALL NOT BE SALVAGED FOR REUSE.

CONSTRUCTION JOINTS:
UNFINISHED WORK SHALL NOT STAND MORE THAN 30 MINUTES UNLESS JOINTS ARE PREPARED. ENTIRE JOINT SURFACE SHALL BE CLEANED, ROUGHENED, AND DAMPENED PRIOR TO APPLICATION OF ADDITIONAL GUNITE.

FINISHING:
BRING GUNITE LAYERS TO WITHIN 1/4 INCH OF FINAL FINISHED SURFACE. WHEN SURFACE HAS TAKEN ITS INITIAL SET, TRIM EXCESS MATERIAL WITH A SHARP EDGE CUTTING SCREED.

CURING
IMMEDIATELY FOLLOWING GUNITE FINISHING, SURFACES SHALL BE CURED FOR NOT LESS THAN SEVEN DAYS USING AN APPROVED CURING METHOD AS SPECIFIED IN ACI 506.2.

INSPECTION:
AFTER GUNITE HAS BEEN IN PLACE FOR A MINIMUM OF 14 DAYS, INSPECT STRUCTURE FOR CRACKING. ALL CRACKS SHOULD BE REPAIRED BY EPOXY INJECTION OR OTHER METHODS APPROVED BY ENGINEER. IF CRACKING IS EXCESSIVE IN THE OPINION OF THE ENGINEER, CORE SAMPLES WILL BE TAKEN AT THE EXPENSE OF THE CONTRACTOR AND THE STRUCTURE EVALUATED BY ENGINEER FOR REPAIRABILITY. IF DEEM UNREPAIRABLE BY ENGINEER, CONTRACTOR SHALL REMOVE POOL SHELL AND REPLACE AT HIS EXPENSE.

POOL EQUIPMENT

QTY	ITEM	MANUFACTURER	MODEL NO.	DESCRIPTION
2	PUMP, FILTRATION	PENTAIR	340032	MODEL EQ75O, 7.5 HP, 354 GPM @ 65 TDH, 4" X 6" FLANGED CONNECTIONS
1	STRAINER BASKET	PENTAIR	357184	EQ SPARE BASKET STRAINER
6	SAND FILTER	PENTAIR	140342	TR140C3, 7.06 SQ FT, 141 GPM @ 20 GPM/FT²
1	SIGHT GLASS	PENTAIR	50901600	INLINE, REPLACEABLE PLEXIGLASS, 1-1/2" SOCKET CONNECTIONS
1	FLOW METER	BLUE-WHITE	F-30600P	HORIZONTAL PIPE MOUNT, 300 SERIES SADDLE PILOT TUBE, 6" SCH 40 IPS PIPE, 170-1,100 GPM
1	CHLORINE EROSION FEEDER	PENTAIR	R171240	HC-3340 RAINBOW SERIES HIGH CAPACITY FEEDER, 40 LB CAPACITY, 1" HARD PIPE, UP TO 332,000 GAL W/ TRICHLOR
1	CHEMICAL FEEDER FLOW RATE INDICATOR	PULSAFEEDER	U8800424	MEC-O-MATIC, 0-35 GPM, FOR HIGH RATE FEEDER WITH 1 INCH HARD PIPING CONNECTIONS
2	MAIN DRAIN SUMP	ASA MANUFACTURING	FBS-50-818-24-8	18" X 18" X 24" FIBERGLASS, 8" SIDE PIPE CONNECTION
2	VGB GRATE COVER & FLANGE	AQUASTAR	18MF101	18" X 18", 147.1 SQ IN, 1,175 GPM @ 2.6 FPS MAXIMUM FLOOR FLOW RATE
2	HYDROSTATIC RELIEF VALVE	HAYWARD	SP1056	1-1/2" THREAD
2	COLLECTOR TUBE	HAYWARD	SP1055	1-1/2" OR 2" MIP
1	AUTOFILL	PENTAIR	T40FW	AUTOMATIC WATER FILLER, SIDE MOUNTED FLOAT VALVE, WHITE LID
22	SURFACE SKIMMER	PENTAIR	506370	BERMUDA MODEL 2" GUNITE SKIMMERS
1	EXTRA SKIMMER BASKET	PENTAIR	516112Z	EXTRA SKIMMER BASKET
6	VACUUM WALL FITTING	HAYWARD	W400BWHP	VAC LOCK SAFETY WALL FITTING X 1-1/2" MPT, WHITE
1	VACUUM ADAPTER: WALL FITTING TO VACUUM HOSE	HAYWARD	AXV092	VACUUM PORT TO 1-1/2" HOSE
6	WALL ADAPTER FITTING	WATERWAY PLASTCS	215-9890	1-1/2" FIPT X 2" SCHEDULE 40 PIPE SPIGOT
36	FLOOR RETURN	STA-RITE	08417-0000	WHITE COLOR, ADJUSTABLE FLOW, NON-CORROSIVE RETURN OUTLET FOR 1-1/2" & 2" PIPE SIZES
9	UNDERWATER POOL LIGHT	PENTAIR	601302	INTELLIBRITE 5G LED, 120 VAC, 55 WATT WHITE (500 EQUIVALENT), 100' CORD
9	LARGE LIGHT NICHE	PENTAIR	78210600	STAINLESS STEEL, 1" REAR HUB, INTERNAL/EXTERNAL GROUND LUGS
9	WET NICHE LUG SEALANT	3M	2135	SCOTCHCAST WET-NICHE POTTING COMPOUND
3	JUNCTION BOX	INTERMATIC	PJB2175	1/2", 3/4", OR 1" PORTS, UP TO 2 LIGHTS, UL LISTED FOR SWIMMING POOLS
1	JUNCTION BOX	INTERMATIC	PJB4175	1/2", 3/4", OR 1" PORTS, UP TO 4 LIGHTS, UL LISTED FOR SWIMMING POOLS
4	HANDRAIL	S.R. SMITH	2HR-4-065	COMMERCIAL STAIR RAIL, 1.9" O.D. X .065" WALL X 48" LONG, 304 STAINLESS STEEL, 2 BEND
1	HANDRAIL	S.R. SMITH	CUSTOM	COMMERCIAL STAIR RAIL, 1.9" O.D. X .065" WALL X 84" LONG, 304 STAINLESS STEEL, 3 BEND
4	LADDER	S.R. SMITH	LF-24-3B	COMMERCIAL (3) NON-SLIP TREADS, 304 STAINLESS STEEL, 1.90" O.D. X .065" WALL X 24" WIDE
4	ANCHOR SET	PERMA-CAST	PC-4020-BC	CHANNEL MOUNTED SET OF 4" BRONZE ANCHORS, 20" CENTERS
10	WEDGE ANCHOR	S.R. SMITH	AS-100B	1.9" X 4" BRONZE SOCKET WITH LOCKING BRONZE WEDGE & STAINLESS STEEL BOLT
18	ESCUTCHEON	S.R. SMITH	EP-100F	4.5" DIAMETER, CHROME PLATED
2	ADA LIFT	S.R. SMITH	575-0000N	MULTILIFT MODEL. ADA COMPLIANT, SELF OPERATING, 350 LB CAPACITY. INCLUDES HANDSET, BATTERY & CHARGER
2	ANCHOR JIG	S.R. WMTH	500-5000A	COMES WITH ANCHOR
2	ADA LIFT ACCESSORY	S.R. SMITH		WHEEL-A-WAY OPTION TO TRANSPORT THE LIFT

LIFE SAVING EQUIPMENT

QTY	ITEM	MANUFACTURER	MODEL NO.	DESCRIPTION
2	UTILITY POLE	PENTAIR	R191106	MODEL 820-12, 12' STRAIGHT ANODIZED POLE
2	LIFE HOOK (SHEPHARD'S CROOK)	PENTAIR	R221026	MODEL 153, ALUMINUM, PERMANENTLY ATTACHED TO UTILITY POLE WITH BRASS SCREWS & WING NUTS
2	HARDWARE KIT	PENTAIR	R221030	MODEL 153, BRASS SCREWS AND WING NUTS
2	LIFE RING	BERKELEY	RC-24W	20" DIAMETER, U.S. COAST GUARD APPROVED 160.050/48/2
2	THROW LINE	BERKELEY	PB504	1/4" X 50' POLYETHYLENE ROPE & FLOAT ATTACHED TO LIFE RING

MAINTENANCE EQUIPMENT

QTY	ITEM	MANUFACTURER	MODEL NO.	DESCRIPTION
1	VACUUM HEAD	PENTAIR	R201126	MODEL 250, 19" SWIVEL WHEEL
1	EXTENSION POLE	PENTAIR	R191306	MODEL 806-16, 2 PIECE ALUMINUM POLE, EXTENDS FROM 8' TO 15.5', E-Z LOCK CAM, BLUE
1	WALL BRUSH	PENTAIR	R111316	MODEL 902, 18" ALUMINUM BACK, WHITE POLYPROPYLENE
1	HAND SKIMMER	PENTAIR	R121196	MODEL 126, HEAVY DUTY ALUMINUM FRAME, REPLACEABLE NET
1	VACUUM HOSE	LAWRENCE	PL50	1.5" X 50' FLOATING SUCTION HOSE
1	WATER TEST SET	TAYLOR CHEMICAL	K2005	DPD, pH, TA, CH, CYA

REQUIRED SIGNS

QTY	ITEM	MANUFACTURER	MODEL NO.	DESCRIPTION
A.R.	NORTH CAROLINA POOL RULES	PENTAIR	R234100	24" X 36", VISIBLE FROM ALL BATHER ENTRANCES, COMPLIES WITH 15A NCAC 18A .2530 (c)(d)
1	WARNING NO LIFEGUARD ON DUTY	PENTAIR	R230500	24" X 18", VISIBLE TO ENTIRE POOL AREA, COMPLIES WITH NCAC 18A .2530 (c)
A.R.	NOTICE, POOL CLOSED	PENTAIR	R231400	12" X 6", ONE PER BATHER ENTRANCE, COMPLIES WITH 18A .2530 (e)
1	EMERGENCY, DIAL 911, POOL PHONE NUMBER, PHYSICAL ADDRESS	MYPOOLSIGNS	S2-1542	INCLUDES EMERGENCY NUMBERS AND SPACE FOR PHYSICAL ADDRESS OF POOL, AND TELEPHONE NUMBER AT SITE COMPLIES WITH NCAC 18A .2530 (f)

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1062 s. Batesville Rd.
Greer, SC 29650
TEL: (864) 417 8720
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TAP ROOT FARMS- MAIN
ENTRY FLETCHER, NC 28732

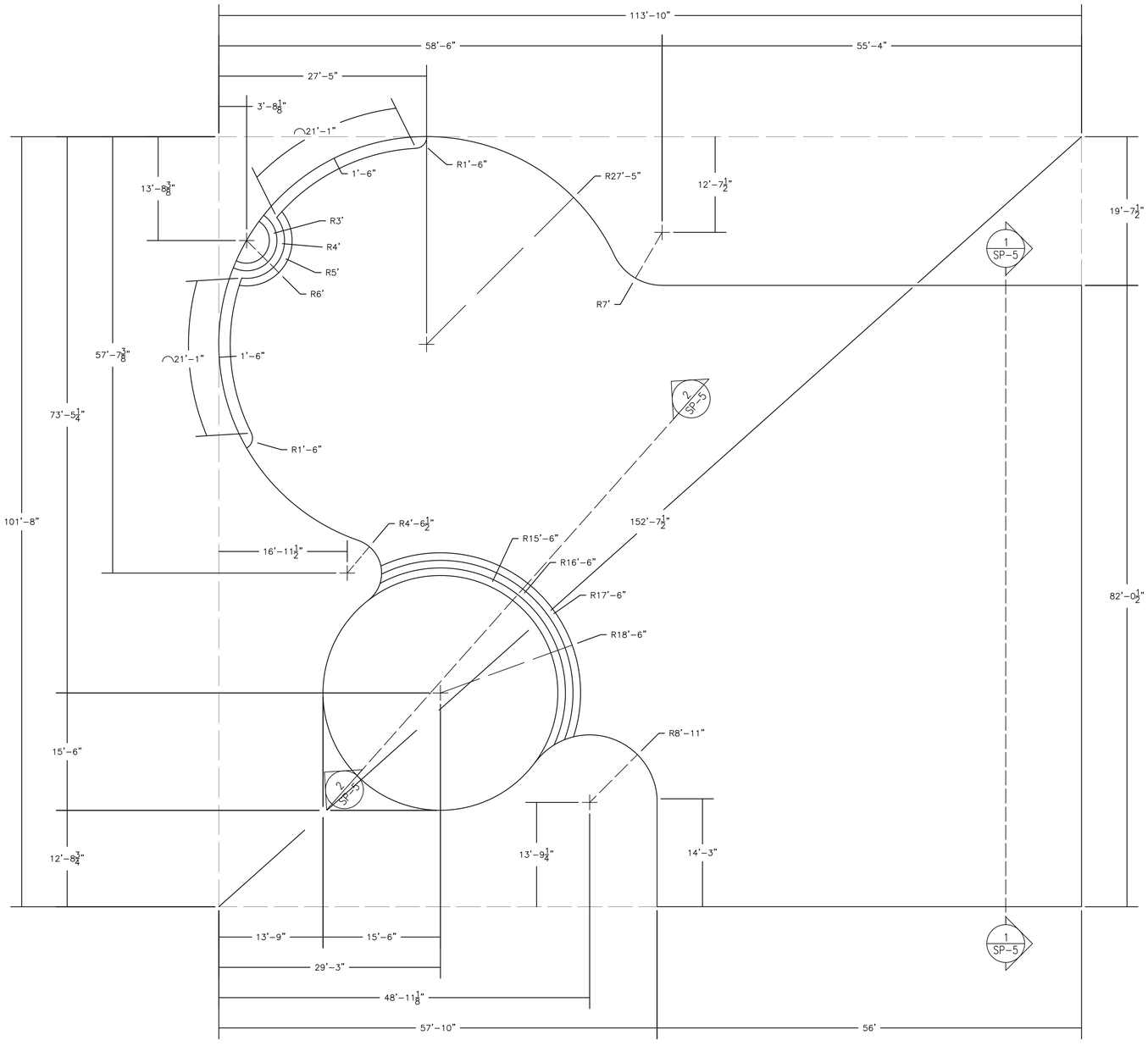
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Drawing Name: PLUMBING LAYOUT

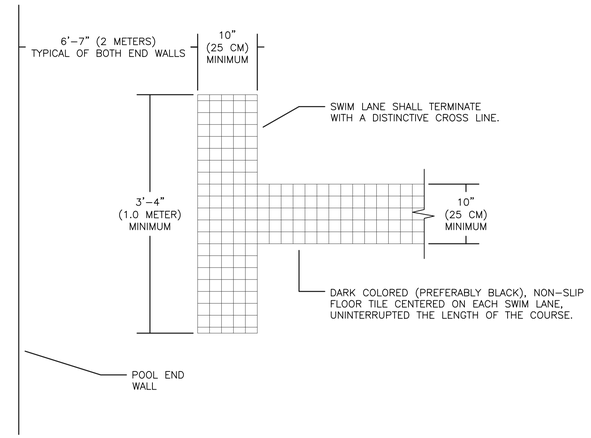
PROJECT NO: 015

DESIGN BY:	PAUL	DRAWING NO.	EM-3
SCALE:	1/4" = 1'-0"		
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00



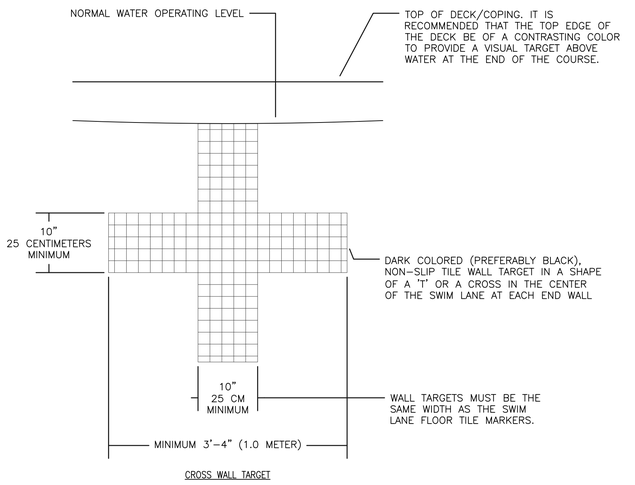
1 SP-3 SCALE: 1/8" = 1'-0"

0' 8' 16' 24'
1/8"=1'-0"



2 SP-3 SCALE: 1"=1'

0' 1' 2' 3'
1" = 1'-0"



CROSS WALL TARGET

3 SP-3 SCALE: 1"=1'

0' 1' 2' 3'
1" = 1'-0"

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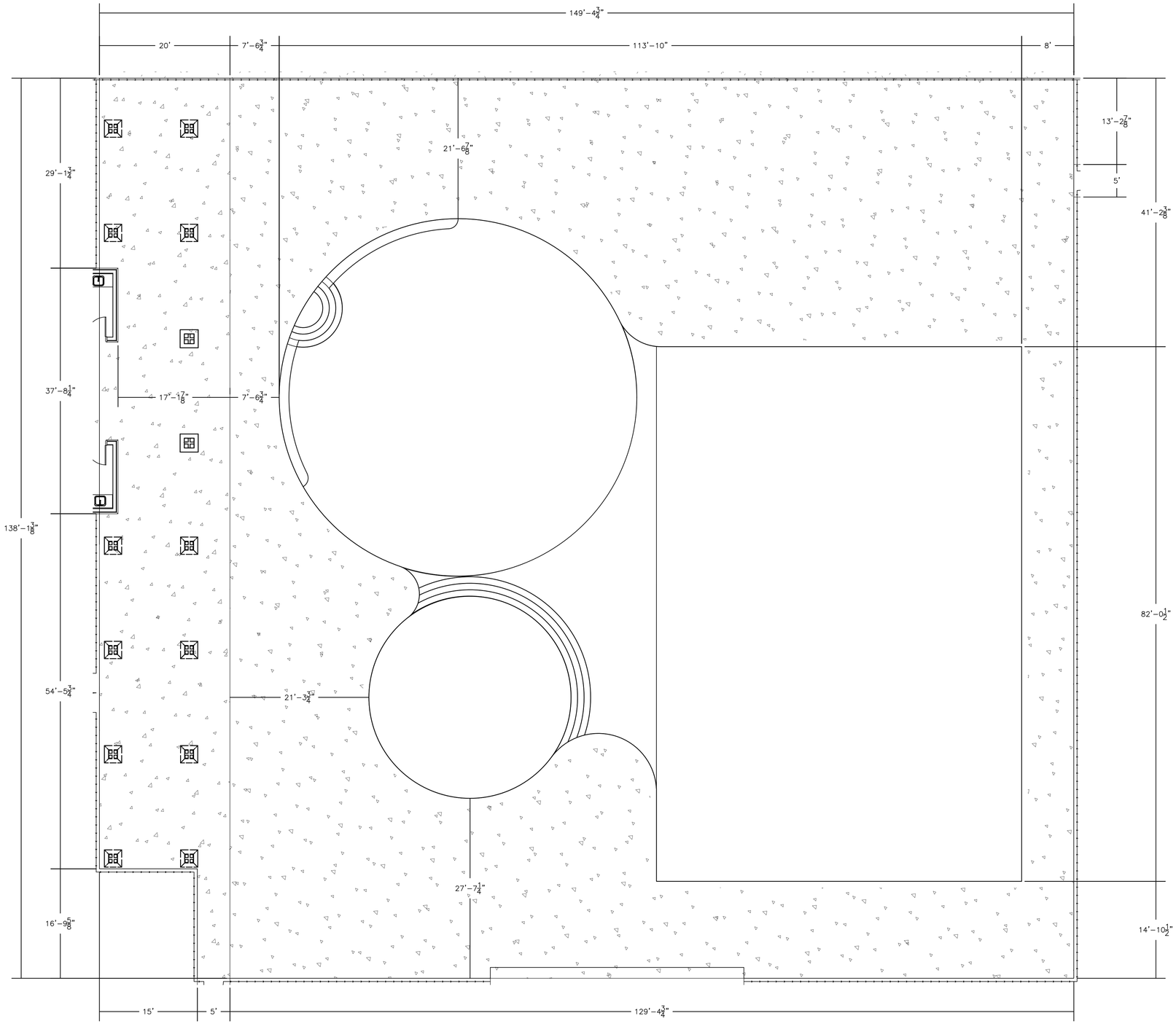
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SEAL:

Drawing Name:
PLUMBING LAYOUT

PROJECT NO:	015

DESIGN BY:	PAUL	DRAWING NO.	
SCALE:	1/4" = 1'-0"	EM-4	
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00



1
SP-4

SCALE: 1/8" = 1'-0"

0' 8' 16' 24'
1/8"=1'-0"

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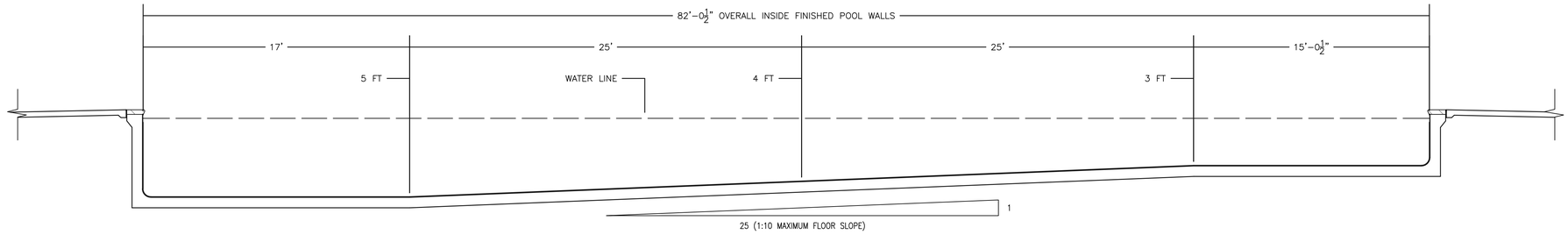
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SEAL:

Drawing Name: PLUMBING LAYOUT

PROJECT NO: 015

DESIGN BY:	PAUL	DRAWING NO.	
SCALE:	1/4" = 1'-0"		EM-5
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00



1 / SP-5 SCALE: 1/4" = 1'-0"

0' 4' 8' 12'
1/4"=1'-0"

2 / SP-5 SCALE: 3/4" = 1'-0"

0' 1' 2' 3'
3/4"=1'-0"

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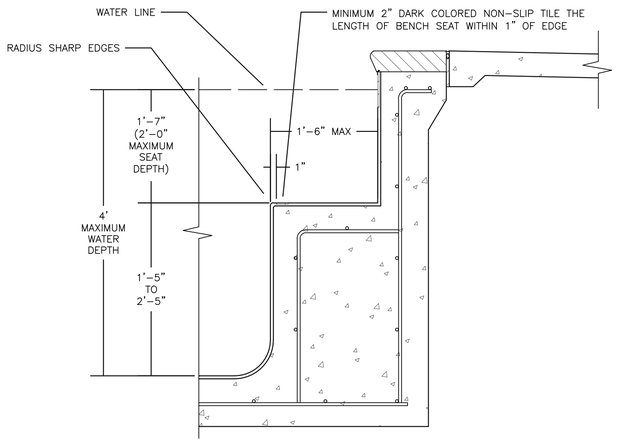
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Drawing Name:
PLUMBING LAYOUT

PROJECT NO:	015

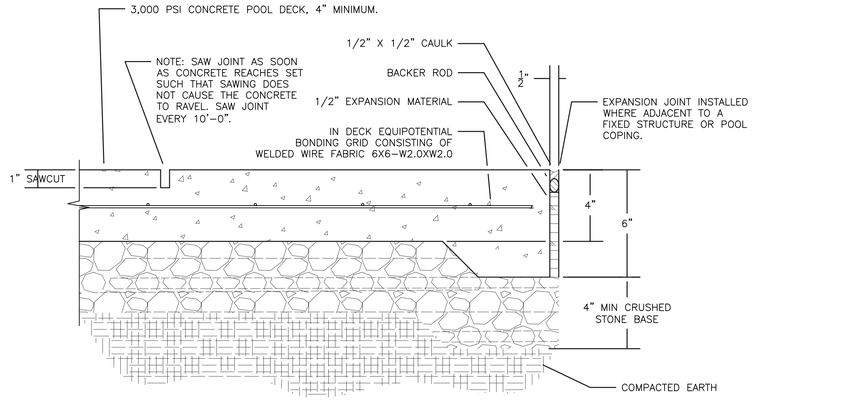
DESIGN BY:	PAUL	DRAWING NO.
SCALE:	1/4" = 1'-0"	EM-6
DRAWN BY:	IVAN	
VERIFY BY:	PAUL	
START DATE:	16 AUG 2023	REVISION # 00



3 / SP-6 SCALE: 1" = 1'-0"

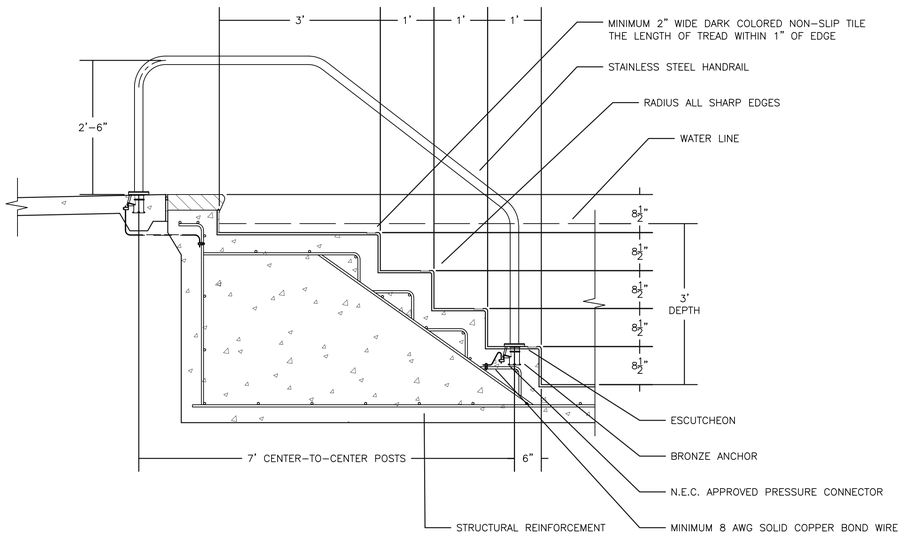
0' 1' 2' 3'
1" = 1'-0"

NOTE: SEE ARCHITECTURAL DRAWINGS FOR DECK TURNDOWN EDGES, SLOPES, MATERIALS, SURFACE FINISHES, AND/OR STRUCTURAL REINFORCEMENT.



4 / SP-6 SCALE: 3" = 1'-0"

0' 1'
3" = 1'-0"



2 / SP-6 SCALE: 3/4" = 1'-0"

0' 1' 2' 3'
3/4"=1'-0"

0 1 2
1" = 1'-0"

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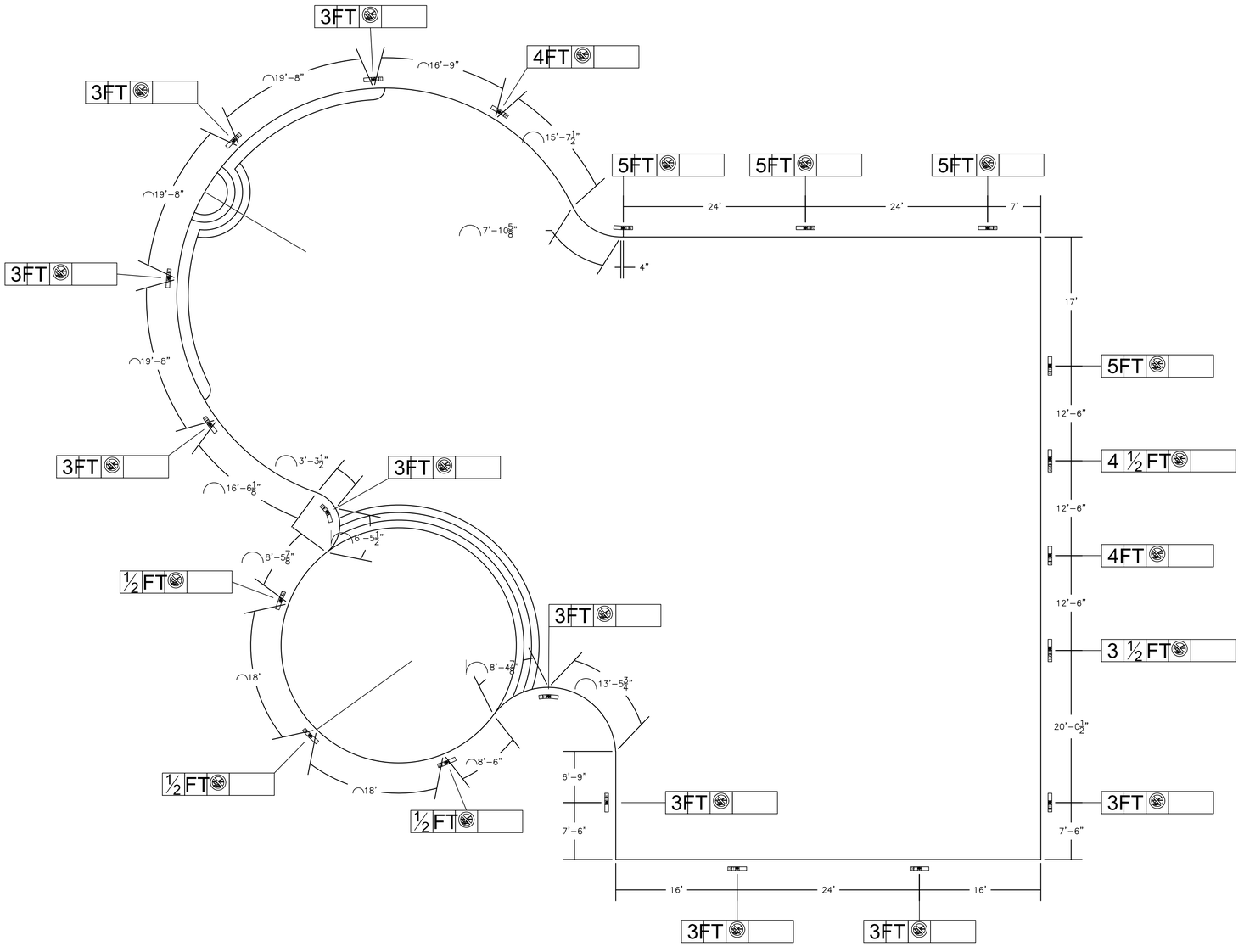
SEAL:

Drawing Name: PLUMBING LAYOUT

PROJECT NO: 015

DESIGN BY:	IVAN	DRAWING NO.	EM-7
SCALE:	N/A		
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00

NOTE:
DEPTH MARKERS MUST BE SPACED AT NOT MORE THAN TWENTY FIVE (25) FOOT INTERVALS ON CENTER, AS MEASURED AROUND THE PERIMETER OF THE POOL.



1
SP-7
SCALE: 1/8" = 1'-0"

0' 8' 16' 24'
1/8"=1'-0"

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SEAL:

Drawing Name:
PLUMBING LAYOUT

PROJECT NO: 015

DESIGN BY:	IVAN	DRAWING NO.	EM-7
SCALE:	N/A		
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00

KEY

MD MD MD MD — POOL MAIN DRAIN SUCTION LINE

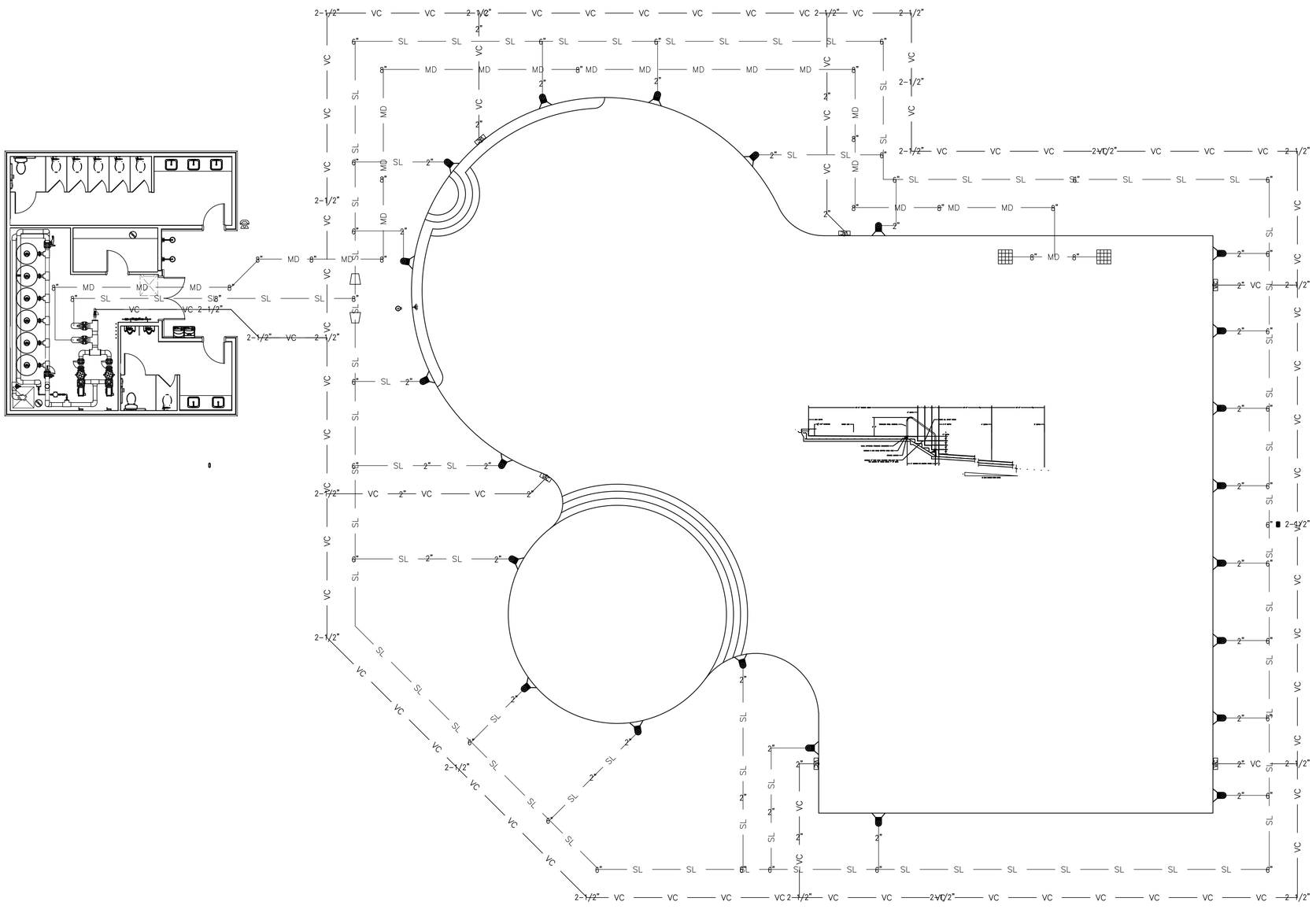
SL SL SL SL — POOL SKIMMER SUCTION LINE

VC VC VC VC — POOL VACUUM SUCTION LINE

NOTE:
PIPING RUNS SHOWN LAID OUT FOR CLARITY & NOT NECESSARILY REQUIRED TO BE INSTALLED EXACTLY LIKE THIS. COMMON SENSE & SITE CONDITIONS MAY DICTATE ACTUAL PLACEMENT.

NOTE:
TO AVOID ENTRAPMENT, ALL SUCTION OUTLETS SHALL BE DESIGNED AND INSTALLED IN ACCORDANCE WITH ANSI/APSP-7.

NOTE:
ALL SCHEDULE 40 PVC PRESSURE RATED PIPING UNLESS NOTED OTHERWISE.
ONLY SDR 26 (CLASS 160) AND SDR 21 (CLASS 200) PVC PIPE MEETING ASTM STANDARD D1785 OR D2241 ARE ACCEPTABLE IN SIZES 12" & SMALLER. NO HEAT BENDING OF PVC PIPE IS ALLOWED.
CPVC PIPE SHALL BE USED FOR CONNECTING TO HEATERS.



1
SP-8

SCALE: 1/8" = 1'-0"

0' 8' 16' 24'
1/8"=1'-0"

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SEAL:

Drawing Name:
PLUMBING LAYOUT

PROJECT NO: 015

DESIGN BY:	IVAN	DRAWING NO.	
SCALE:	N/A	EM-7	
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00

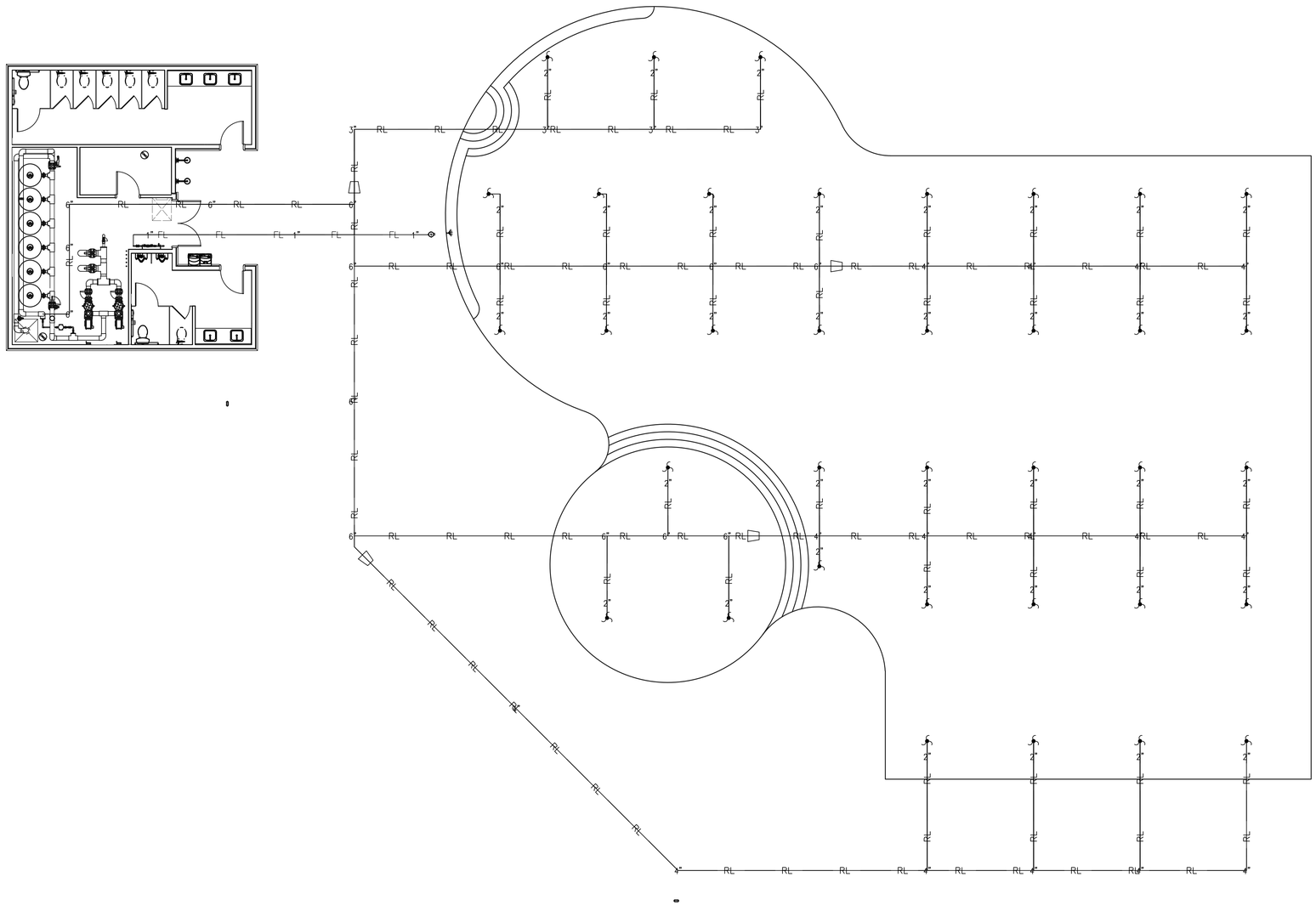
KEY

RL RL RL RL FILTRATION RETURN LINE

FL FL FL FL AUTOFILL LINE

NOTE:
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NOTE:
ALL SCHEDULE 40 PVC PRESSURE RATED PIPING UNLESS NOTED OTHERWISE.
ONLY SDR 26 (CLASS 160) AND SDR 21 (CLASS 200) PVC PIPE MEETING ASTM STANDARD D1785 OR D2241 ARE ACCEPTABLE IN SIZES 12" & SMALLER. NO HEAT BENDING OF PVC PIPE IS ALLOWED. CPVC PIPE SHALL BE USED FOR CONNECTING TO HEATERS.



1 / SP-9 SCALE: 1/8" = 1'-0"

0' 8' 16' 24'
1/8"=1'-0"

Carolina Pools LLC
1062 s. Batesville Rd.
Greer, SC 29650
TEL: (864) 417 8720
carolinapoolsllc.com

CLIENT
DR.HORTON

TAP ROOT FARMS- MAIN
ENTRY FLETCHER, NC 28732

NO.	EMISSIONS & REVISIONS	DATES
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SEAL:

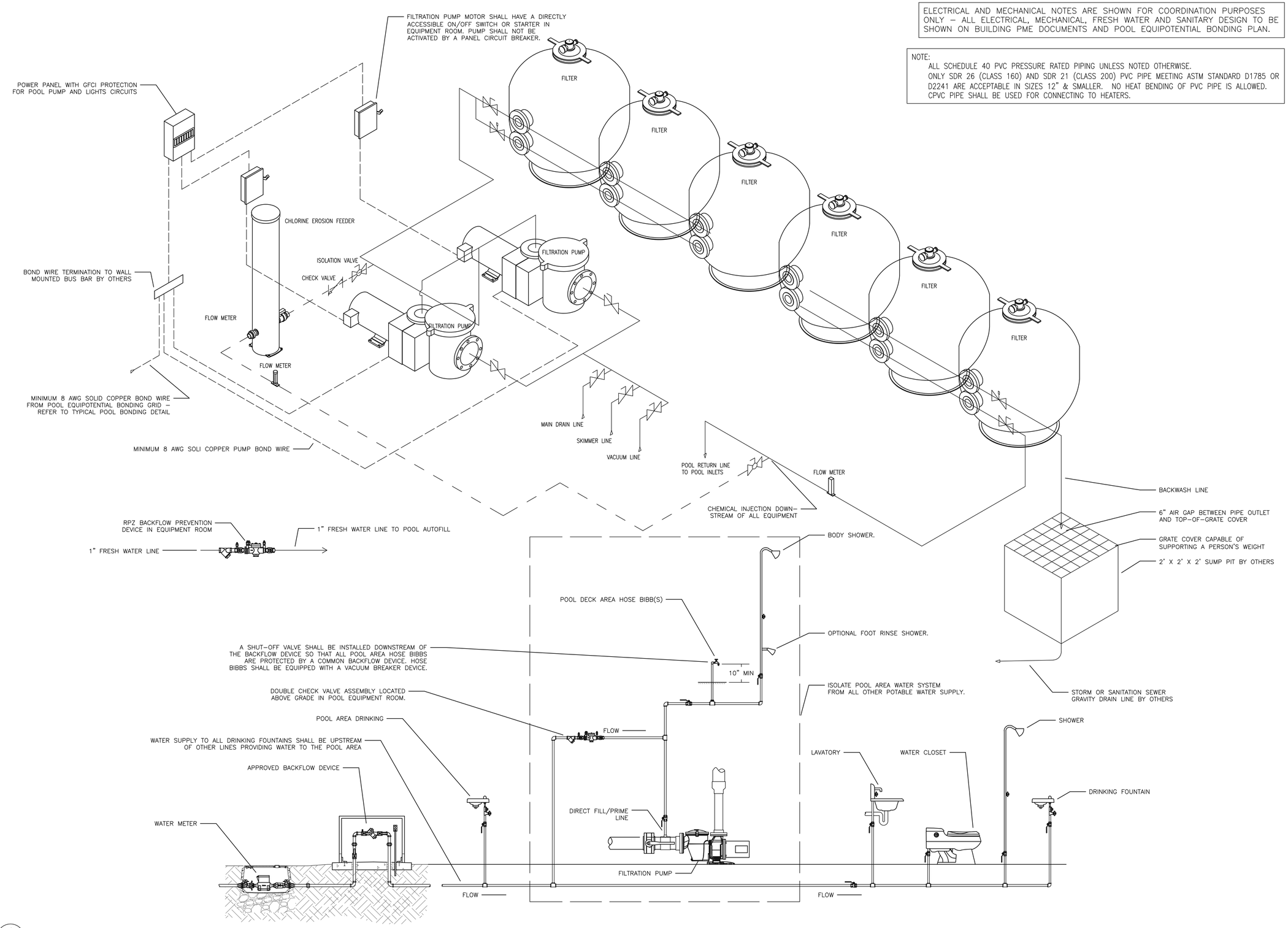
Drawing Name:
PLUMBING LAYOUT

PROJECT NO: 015

DESIGN BY:	IVAN	DRAWING NO.	
SCALE:	1/8" = 1'0"		SP-10
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00

ELECTRICAL AND MECHANICAL NOTES ARE SHOWN FOR COORDINATION PURPOSES ONLY – ALL ELECTRICAL, MECHANICAL, FRESH WATER AND SANITARY DESIGN TO BE SHOWN ON BUILDING PME DOCUMENTS AND POOL EQUIPOTENTIAL BONDING PLAN.

NOTE:
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CPVC PIPE SHALL BE USED FOR CONNECTING TO HEATERS.



1
SP–10
SCALE: NTS

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ENTRY FLETCHER, NC 28732

NO.	EMISSIONS & REVISIONS	DATES
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SEAL:

Drawing Name:
PLUMBING LAYOUT

PROJECT NO: 015

DESIGN BY:	IVAN	DRAWING NO.	
SCALE:	N/A		SP-11
DRAWN BY:	IVAN		
VERIFY BY:	PAUL		
START DATE:	16 AUG 2023	REVISION #	00